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The molecular basis of heat stress responses in plants

Yi Kan[1,4], Xiao-Rui Mu[1,3], Jin Gao[1,3], Hong-Xuan Lin[1,3,4,*] and Youshun Lin[2,*]

[1]National Key Laboratory of Plant Molecular Genetics, CAS Centre for Excellence in Molecular Plant Sciences, Shanghai Institute of Plant Physiology and Ecology, Chinese Academy of Sciences, Shanghai 200032, China

[2]Joint Center for Single Cell Biology, School of Agriculture and Biology, Shanghai Jiao Tong University, Shanghai 200240, China

[3]University of the Chinese Academy of Sciences, Beijing 100049, China

[4]Guangdong Laboratory for Lingnan Modern Agriculture, Guangzhou 510642, China

***Correspondence: Hong-Xuan Lin (hxlin@cemps.ac.cn), Youshun Lin (linyoushun@sjtu.edu.cn)**

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ABSTRACT

Global warming impacts crop production and threatens food security. Elevated temperatures are sensed by different cell components. Temperature increases are classified as either mild warm temperatures or excessively hot temperatures, which are perceived by distinct signaling pathways in plants. Warm temperatures induce thermomorphogenesis, while high-temperature stress triggers heat acclimation and has destructive effects on plant growth and development. In this review, we systematically summarize the heat-responsive genetic networks in *Arabidopsis* and crop plants based on recent studies. In addition, we highlight the strategies used to improve grain yield under heat stress from a source–sink perspective. We also discuss the remaining issues regarding the characteristics of thermosensors and the urgency required to explore the basis of acclimation under multifactorial stress combination.

Key words: global warming, thermosensing, heat stress, thermomorphogenesis, heat stress responses, heat acclimation

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INTRODUCTION

Agricultural harvests are closely related to environmental temperature. Recently, as a result of global warming, the frequent occurrence of extremely high temperatures has led to crop yield losses and widespread threats to food security and agricultural sustainability, raising a range of potential social problems (Challinor et al., 2014). As sessile organisms, plants are unable to move to favorable environments when they experience high temperatures. Understanding the molecular basis of how plants respond to and tolerate high temperatures, especially in crops, is therefore critical if policymakers, agriculturalists, and crop breeders are to address the global food security issues caused by global warming.

Not all temperature increases are harmful to plants. Heat waves, heat shock, and warming affect plants to varying degrees and in different ways (Jagadish et al., 2021). Mild or moderate temperature increases trigger a developmental phase transition called thermomorphogenesis, which includes hypocotyl elongation, flowering alteration, and other morphological and architectural changes (Casal and Balasubramanian, 2019). The regulatory network in thermomorphogenesis, including the upper-stream sensors, the center regulators, and the downstream responders, has been well established in *Arabidopsis* (Ding et al., 2020; Kerbler and Wigge, 2023). The upper limit of the mild warm temperature category is defined based on different species; it is up to 27°C for *Arabidopsis* and up to 34°C for rice (Wahid, 2007). Unlike exposure to warm temperatures, heat stress (HS) always has a negative impact on plant growth and development and can even result in cell damage and death. Indeed, agronomic crop production is highly vulnerable to substantial increases in temperature, which can cause catastrophic crop yield losses (Challinor et al., 2014). From a molecular perspective, the photosynthetic efficiency is easily disturbed by HS, contributing to a shortened life cycle and reduced yields; respiratory enzyme activities decline under HS, contributing to energy deficiency; and heat-triggered movement of biomolecules disrupts the homeostasis of the plasma membrane and strongly alters its permeability and

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fluidity, followed by observed leakage of ions and amino acids. In general, when a plant's optimal growth temperature is exceeded by 5°C–10°C, a reactive oxygen species (ROS) burst occurs globally and causes irreversible cellular oxidative damage, especially in photosynthesis systems I and II (Pucciariello et al., 2012; Zhao et al., 2021a). From the perspective of HS-induced plant phenotypes, growth retardation and wilted plant architecture are commonly observed at the vegetative stage, and low pollen viability, abnormal fertilization, and poor grain filling appear at the reproductive stage (Zhao et al., 2021a). To acquire heat acclimation, plants have evolved sophisticated and systemic mechanisms to cope with the effects of HS.

Thermosensing is the first step by which plants respond to increases in temperature by transducing extracellular temperature signals into intracellular physiological responses or by directly sensing intracellular heat waves and altering their RNA and protein levels (Kan and Lin, 2021). Thermosensors are a major research focus and have been well characterized in recent studies. Multiple molecular mechanisms have been discovered to explain how plants sense elevated temperatures, including phase separation, isomerization, subcellular translocation, RNA structure switches, and chromatin remodeling (Kumar and Wigge, 2010; Jung et al., 2016, 2020; Chung et al., 2020). However, thermosensing in crops under HS is not well understood at present. A recent study showed that TT3.1 (Thermos-tolerance 3.1) functions as a potential thermosensor under HS conditions and acts to transduce temperature signals from the plasma membrane to the chloroplast (Zhang et al., 2022a). The results of this initiated the exploration for crop thermosensors in future investigations.

The systemic HS response regulatory network has been progressively established by studies in *Arabidopsis* (Ding et al., 2020). HSFs (heat shock factors) and HSPs (heat shock proteins) play dominant roles in activating HSRs (heat shock responses), and some heat-responsive signal molecules (e.g., Ca^{2+}, NO, hormones) act upstream of HSF. HSR gene expression is also under the control of non-coding RNAs (ncRNAs) and epigenetic modulation; the reaction of UPRs (unfolded protein responses) in the ER (endoplasmic reticulum) occupies an important position in HS responses. Mitochondria and chloroplasts, which are thermosensitive organelles, deploy diverse approaches to cope with HS. Furthermore, there is a growing understanding of HS responses in crop plants, especially in rice, where three major thermotolerance quantitative trait loci, also known as TT1, TT2, and TT3, have been identified that can significantly improve grain yield under HS conditions (Li et al., 2015; Zhang et al., 2022a; Kan et al., 2022). As the detailed molecular mechanisms are discovered, the HS response regulatory networks are being mapped out in crop species.

The overview of temperature responses mentioned above is essential for understanding how plants acquire heat acclimation. Because the current understanding of thermosensing and thermal responses is rather limited, our focus in this review is on summarizing and discussing recent findings regarding the molecular regulatory mechanisms of thermosensing, the differences between thermomorphogenesis and heat damage, thermal responses, and the strategies for heat resilience. In addition, we provide some perspectives on the remaining questions.

GLOBAL WARMING THREATENS FOOD SECURITY

Global warming has become a major threat to food security and human development. Average global temperatures have increased by 1.25°C as a result of intensifying human activity, and authoritative estimates of the current emissions trajectory suggest that the temperature rise will exceed 1.5°C in less than 10 years (Matthews and Wynes, 2022). For a regional perspective, by 2021, the average temperature change had exceeded 1.5°C in several countries, among which China is particularly notable (Figure 1A). The dynamics of temperature change over the meteorological year and in the summer season around the world and in China from 1961–2021 indicate that global warming is becoming a serious issue (Figure 1B). It has been reported that for every 1°C increase in global temperature, yields of the 4 major crops (wheat, rice, maize, and soybean) are reduced by approximately 6.0%, 3.2%, 7.4%, and 3.1% on average, respectively (Zhao et al., 2017). Furthermore, given the prevalence of moderate or severe food insecurity and the increase in the world population from 2014 to 2021, improving heat tolerance and acclimation in crops is extremely urgent (Figure 1C and 1D).

Rice (*Oryza sativa*), one of the world's most important food crops, provides most of the daily dietary calories for more than 4 billion people. China and India, the world's most populous countries that lead the world in both rice production and in the rice area harvested, are geographically located in tropical and temperate regions of the world and are severely threatened by global warming (Figure 1A, 1E, and 1F). Therefore, elucidating the fundamental mechanisms of plant responses to elevated temperature and mining genetic alleles (genes) from germplasm resources and applying them to the breeding of heat-acclimated crops via gene introgression or genome editing are all indispensable aspects of plant science research to increase global food security in this era of climate change.

THERMAL SENSING AND RESPONSES IN THERMOMORPHOGENESIS

How plants sense warm temperature

Recent studies have shown that phytochromes, the canonical photoreceptors, also function in thermosensing and that active isoforms can spontaneously convert to inactive isoforms in response to elevated temperatures (Jung et al., 2016; Legris et al., 2016; Chen et al., 2022). For example, the well-characterized phyB (Phytochrome B) is a red:far red (R:FR)-sensitive photoreceptor (Jung et al., 2016; Legris et al., 2016; Chen et al., 2022). Red light converts Pr (the inactive form of phyB) into a Pfr (the active form of phyB) homodimer (Cheng et al., 2021). The homodimer, which is localized to the nucleus, subsequently suppresses the activity of the basic-helix-loop-helix transcription factors PIF4 (PHYTOCHROME-INTERACTING FACTOR 4) and PIF7, which function as two master regulators in thermomorphogenesis (Koini et al., 2009; Kumar et al., 2012); elevated temperature facilitates Pfr-to-Pr reversion, removing the inhibition of the Pfr dimer on PIF4 and PIF7 and subsequently activating downstream responses (Figure 2; Jung et al., 2016). There is increasing evidence to suggest that some photosensors may also have a dual function that involves thermosensing, which offers new clues

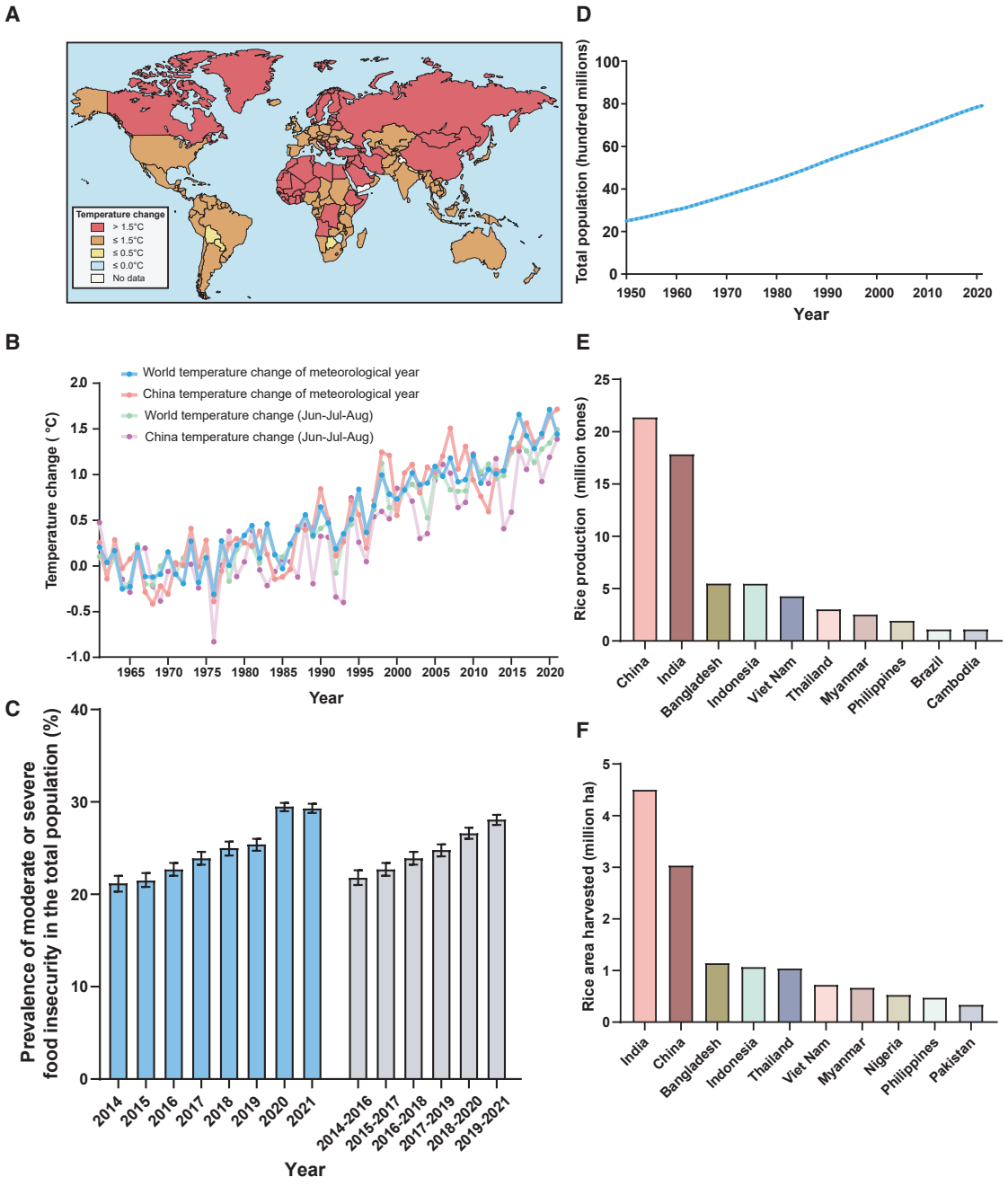


Figure 1. Global warming and rapid population growth threaten food security around the world.

(A) Mean temperature change in different countries in 2021.

(B) Mean temperature change of meteorological years 1961–2021.

(C) The proportion of moderate or severe food insecurity in the total population. The blue columns indicate the annual values. The gray columns indicate 3-year average values. Values are shown as means and confidence intervals.

(D) Population dynamics from 1950–2021.

(E) Top 10 rice-producing countries in terms of total production.

(F) Top 10 rice producers in terms of area harvested.

All data were collected from the Food and Agriculture Organization of the United Nations (www.fao.org).

regarding the identification of thermosensors in crops (Jung et al., 2016; Legris et al., 2016; Chen et al., 2022).

In addition to isomerization, protein phase separation also plays an indispensable role in thermosensing. ELF3 (EARLY FLOWERING 3), one of the components of the evening complex, contains a prion-like domain with abundant glutamine residues and functions to inhibit *PIF4* mRNA levels. During periods of elevated temperature, condensation of ELF3 proteins is observed to form speckles in a prion-like domain-dependent manner, and

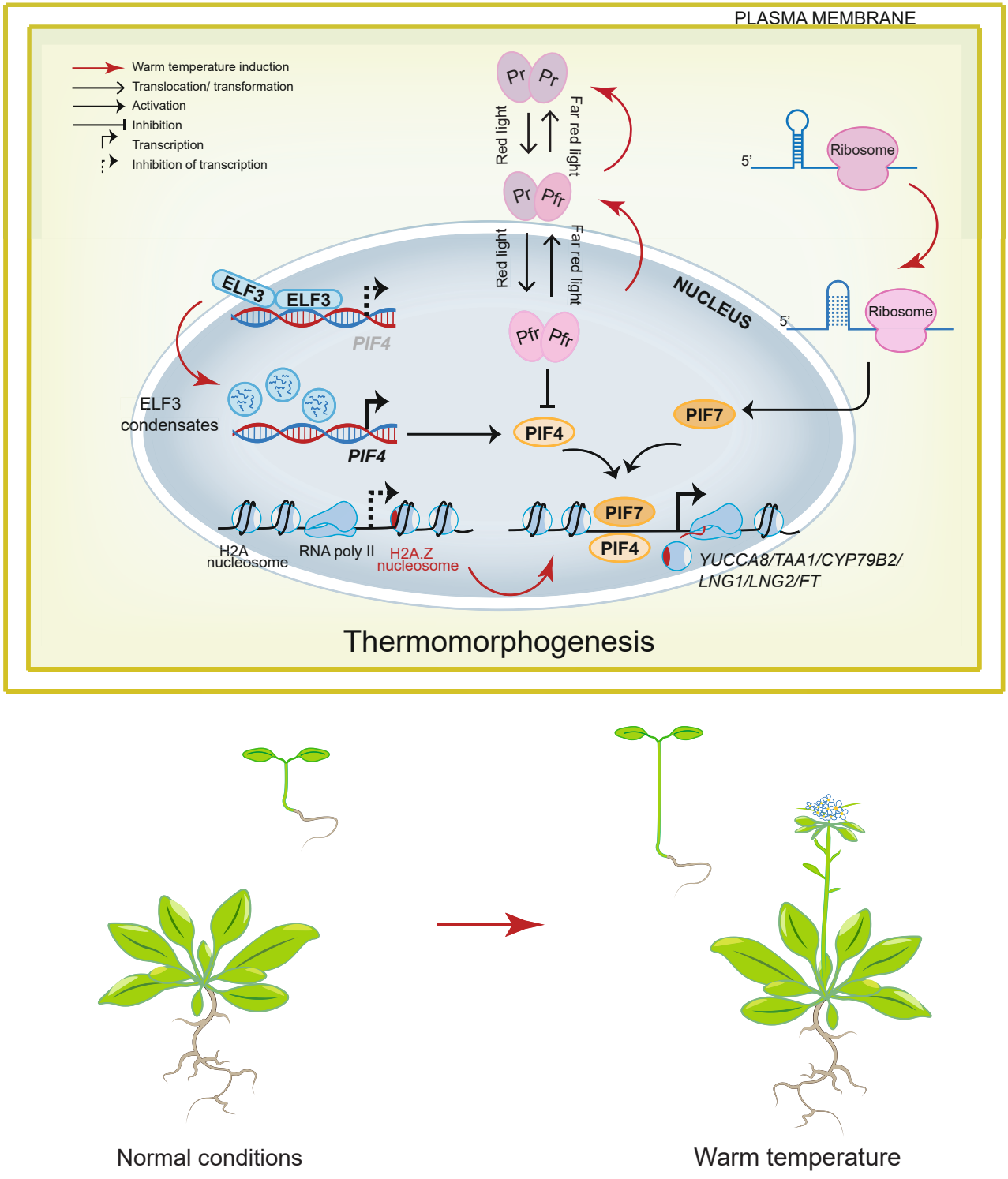


Figure 2. Perception and responses to warm temperature in plants.
Temperature elevation activates PIF4 via the isomerization of phyB and the phase separation of ELF3; RNA structure also functions as a thermal switch to promote PIF7 translation. PIF4 and PIF7 may function upstream of the heat-triggered removal of H2A.Z nucleosomes, eventually initiating thermomorphogenesis-related gene expression, such as *YUCCA8*, *TAA1*, *CYP79B2*, *LNG1*, *LNG2*, *FT*, etc.

this process is reversible and spontaneous, sequestering ELF3 from the *PIF4* promoter and subsequently activating the expression of *PIF4* (Figure 2; Jung et al., 2020).

Previous studies have demonstrated that the mRNA secondary structure is susceptible to environmental temperature in animals and bacteria (Vu et al., 2019). In plants, elevated temperature

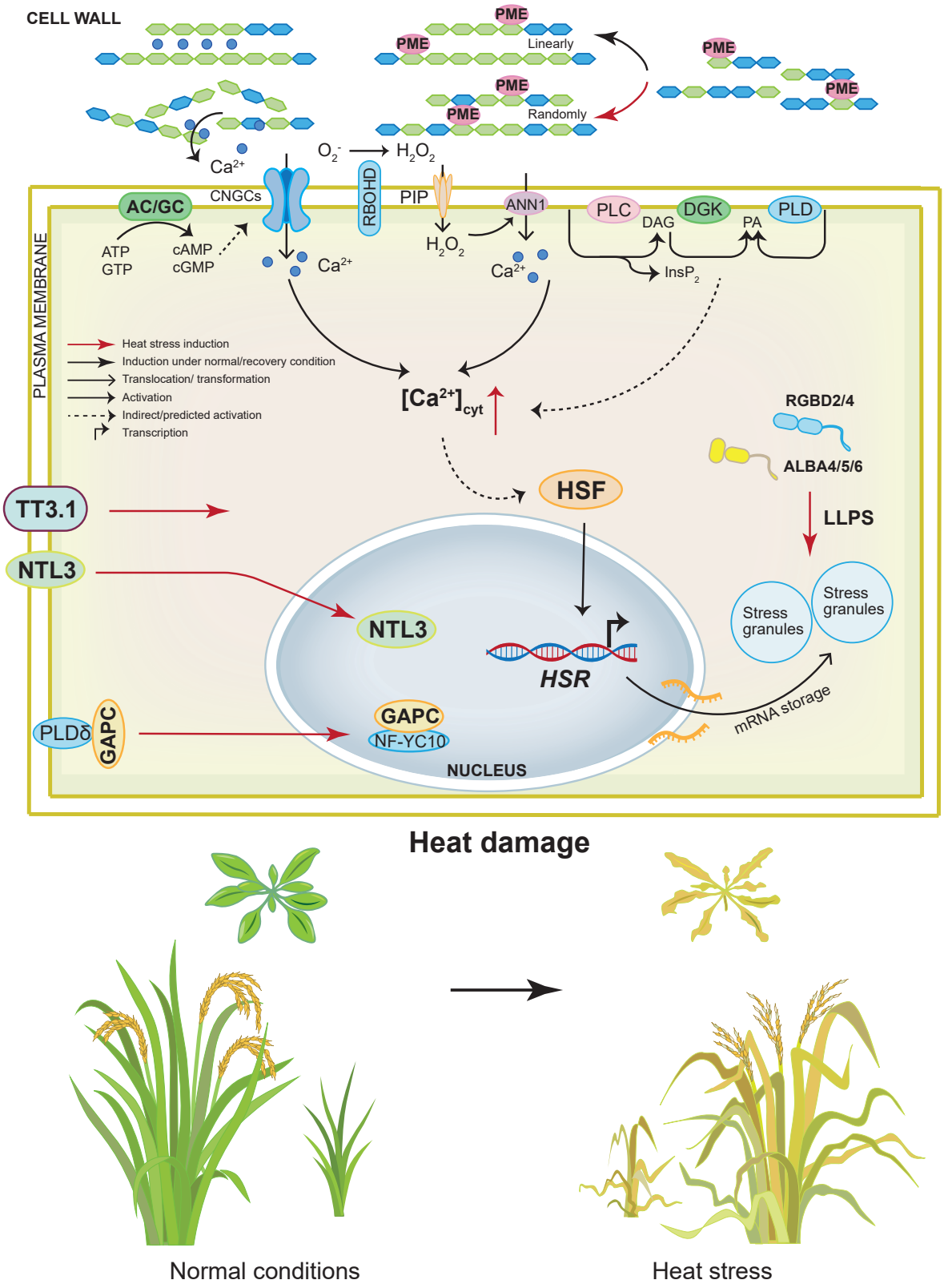


Figure 3. Putative thermosensing upon HS in plants.

Temperature increases trigger cell wall remodeling via PMEs (pectin methylesterases), causing Ca^{2+} influx from the apoplast into the cytoplasm. Because of its structural and compositional specificities, the plasma membrane exhibits thermosensitivity in fluidity, where several second messengers, including

(legend continued on next page)

induces the generation of a hairpin structure in the 5′ UTR of *PIF7* mRNA just upstream of the AUG, promoting *PIF7* translation. The hairpin structure exhibits different configurations depending on ambient temperature (Figure 2) (Chung et al., 2020). Similar hairpin sequences have also been found in *WRKY22*, a gene that responds to fluctuating temperatures, and in the essential heat shock regulator HSFA2, indicating that thermosensing by mRNA structure exists widely in plants and warrants further experimental exploration (Chung et al., 2020).

The molecular responses in thermomorphogenesis

Previous studies have suggested that the alternative histone H2A.Z in the nucleosome plays an indispensable role in sensing environmental temperature. Under normal conditions, H2A.Z-containing nucleosomes have a higher degree of wrapping in local DNA than canonical H2A-containing nucleosomes and largely occupy it, blocking transcription either by directly impeding the function of RNA polymerase II (RNA Pol II) or by sequestering transcription factors from *cis* elements. Elevated temperatures decrease the occupancy of H2A.Z nucleosomes, leading to unobstructed progression of RNA Pol II and the exposure of *cis* elements, which ultimately activate transcription and induce the expression of heat-responsive genes, such as *HSF1* (Kumar and Wigge, 2010; Cortijo et al., 2017). Warm-inducible H2A.Z dynamic also facilitates PIF4 binding to the FLOWERING LOCUS T promoter and activates its transcription (Kumar et al., 2012).

However, recent studies have demonstrated that the low R:FR-triggered removal of H2A.Z occurs downstream of PIF DNA binding (Willige et al., 2021). Dephosphorylation was detected in PIF7 protein when exposed to low R:FR, and subsequently PIF7 got access to bind to its target sites, where PIF7 triggered the removal of H2A.Z via potential interaction with the EIN6 ENHANCER subunit of the INO80 complex and an unknown histone acetyltransferase, eventually activating downstream genes (Figure 2; Willige et al., 2021). Although this process took place upon low R:FR light but not temperature alteration, these results still hint at the possibility that the chromatin remodelers may not function upstream of temperature perception; further experimental investigation needs to be done before reaching a conclusion.

In addition, alternative splicing of the floral repressor FLOWERING LOCUS M mRNA, dimer/monomer photo-equilibrium of the ultraviolet B radiation photoreceptor UV RESISTANCE LOCUS 8, and altered expression of small/long noncoding RNAs also function in thermomorphogenesis regulation (Posé et al., 2013; Findlay and Jenkins, 2016; Sureshkumar et al., 2016; Gyula et al., 2018; Severing et al., 2018). At warm temperatures, PIF4, a helix-loop-helix transcription factor, enhances the transcription of genes such as *YUCCA8*, *TAA1* (*TRYPTOPHAN AMINOTRANSFERASE OF ARABIDOPSIS 1*), *CYP79B2* (*CYTOCHROME P450 FAMILY 79B*), *LNG1* (LONGIFOLIA 1), *LNG2*, and *FLOWERING LOCUS T*, which are involved in plant stem growth and flowering, while decreasing the expression of *SPCH* (*SPEECHLESS*), a gene involved in stomatal development (Figure 2) (Casal and Balasubramanian, 2019). In addition to PIF4- and phyB-mediated thermomorphogenesis, TOT3 (TARGET OF TEMPERATURE 3), encoding a plasma membrane-localized kinase, MAP4K4 (MITOGEN-ACTIVATED PROTEIN KINASE KINASE KINASE KINASE 4), is also implicated in the regulation of thermomorphogenesis in *Arabidopsis* and wheat (Vu et al., 2021).

PUTATIVE THERMOSENSING UPON HS

Unlike warm temperature sensing, the thermosensing upon HS remains largely enigmatic. Thermosensors, broadly defined, are proteins capable of decoding elevated temperature as a cellular signal by altering their own structures and/or activities, by activating second messengers, or by interacting with other molecular components (Vu et al., 2019; Kan and Lin, 2021; Kerbler and Wigge, 2023). We here describe some putative thermosensors that respond to HS in plants.

Cell wall remodeling releases Ca^{2+} in response to heat

The cell wall is the first protective barrier in plants because it is exposed to heat prior to the plasma membrane. Under ambient environmental conditions, apoplastic Ca^{2+} is sequestered among de-esterified pectin residues. Moderate HS induces PME (pectin methylesterases) activity randomly on the pectin, where protons are released by demethylesterification, lowering the pH of the cell wall. Endo-polygalacturonase is subsequently activated, causing cell wall loosening and Ca^{2+} mobilization from apoplasts (Figure 3; Wu and Jinn, 2010). Ca^{2+} flows into the cytoplasm through potential Ca^{2+}-permeable channels, ultimately activating the HSR (Wu and Jinn, 2010; Wu et al., 2018).

Chemical messengers are activated in the plasma membrane

Biological membranes, which consist of highly ordered lipids and proteins, are the most thermosensitive cell components of living organisms (Niu and Xiang, 2018). Elevated temperature causes changes in lipid composition, disordered fatty acid configuration, and disruption of rotation, diffusion, and head group packing density of lipid bilayers. Collectively, these alterations are referred to as membrane fluidity changes, and they have direct or indirect effects on the folding, mobility, and activity of membrane proteins (Kerbler and Wigge, 2023). Considering that membrane fluidity is sensitive to elevated temperature, the plasma membrane has the foundation to perform a thermosensing function (Hayes et al., 2021). The plasma membrane separates the apoplast from the cytoplasm; therefore, some membrane proteins, which acquire the ability to translate signals from apoplastic mobile chemical messengers into elevated cytoplasmic calcium ion levels, are

Ca^{2+}, cAMP/cGMP, ROS, and signal lipids, are activated to transduce temperature signals into downstream physiological responses. Ca^{2+} enters the cell interior via Ca^{2+}-permeable channels and other associated proteins, such as CNGCs and ANNs; CNGCs are activated by cAMP/cGMP, and the activation of ANNs relies on H_2O_2 (a major type of ROS). However, in turn, Ca^{2+} also functions upstream of ROS generation. Phospholipid PAs generated by PLD and PLC/DAG kinase (DGK) function as signal molecules by targeting downstream proteins. Inositol phosphates hydrolyzed by PLCs trigger Ca^{2+} release. RGBD2/4 and ALBA4/5/6 sense HS to form SGs (stress granules) via phase separation and protect HSR-related mRNA. Heat-dependent nuclear translocation of GAPC and NTL3 activates the expression of downstream genes. In addition, TT3.1, which relocates from the plasma membrane to the endosome in response to HS, also functions as a putative thermosensor.

regarded as potential thermosensors in plants. CNGCs (cyclic nucleotide-gated channels) facilitate heat-triggered Ca^{2+} influx in the moss *Physcomitrella patens*, *Arabidopsis*, and rice; however, whether CNGCs can function as thermosensors to spontaneously respond to HS in a manner similar to osmosensors (i.e., the hyperosmolality-gated calcium-permeable channel AtOSCA1), will require further investigation (Figure 3; Saidi et al., 2009; Finka et al., 2012; Yuan et al., 2014; Cui et al., 2020). Nevertheless, two common chemical molecules, cyclic AMP (cAMP) and cyclic guanosine monophosphate (cGMP), are generated by adenylyl/guanylate cyclase or by the Toll/interleukin-1 receptor (Yang et al., 2021; Yu et al., 2022). Given that numerous cAMP-induced proteins overlap with those that are expressed in a temperature-responsive manner, along with the activation of CNGCs by cAMP/cGMP, we speculate that the membrane-localized cAMP/cGMP producers may work upstream of CNGCs and function as potential thermosensors in plants (Figure 3; Thomas et al., 2013; Lu et al., 2016; Gehring and Turek, 2017; Yang et al., 2021). In addition, ANN1 (annexin 1) has been shown to be required for heat-triggered Ca^{2+} level increase, and further studies demonstrate that it is activated by H_2O_2 and suppressed by MYB30 at the mRNA level, raising a question about whether Ca^{2+} or H_2O_2 works upstream (Figure 3; Richards et al., 2014; Liao et al., 2017). A well-characterized model has been established based on a dual calcium/ROS mechanism in salt stress responses under the control of RBOHD (respiratory burst oxidase homolog D) and calcium channels, in which there is crosstalk between Ca^{2+} and ROS, ultimately generating self-propagating Ca^{2+}/ROS waves. In response to HS, RBOHD is activated at high temperatures and causes the accumulation of ROS. ANNs function downstream to sense ROS and promote Ca^{2+} influx. Elevated Ca^{2+} levels in the cytoplasm function to activate RBOHD, which, in turn, may enable temperature sensing in plants (Figure 3; Miller et al., 2009; Evans et al., 2016; Zhu, 2016).

Apart from Ca^{2+} and ROS, lipids also function as signal molecules to transduce environmental cues into intracellular responses. PAs (phosphatidic acids) generated by PLD (phospholipase D) and PLC/DGK (phospholipase C/diacylglycerol [DAG] kinase) respond rapidly to HS within minutes and exhibit dynamic fluctuation (Mishkind et al., 2009). The increased proportion of negatively charged PA destabilizes the plasma membrane, altering the activities of membrane proteins. In addition, PA can also directly bind to downstream targeted proteins and activate or inhibit downstream responses, indicating that the generation of these lipids is closely tied to thermal perception (Hou et al., 2016). Despite the fact that the heat-triggered rapid PA responses are largely dependent on PLD, the activation of PLD requires Ca^{2+} and H_2O_2, suggesting that PLD functions downstream of the primary thermosensing (Hayes et al., 2021). Phosphatidylinositol 4,5-bisphosphate and phosphatidylinositol 4-phosphate are hydrolyzed by PLCs to generate DAG and inositol-1,4,5-trisphosphate. Inositol-1,4,5-trisphosphate in the cytoplasm may trigger Ca^{2+} release from intracellular compartments, conferring membrane localization on PLCs, which may play a role in thermosensing (Figure 3; Munnik et al., 1998; Munnik, 2014; Rupwate and Rajasekharan, 2012). The PLCs are classically activated by some receptor kinases and G proteins in animals; considering that G protein–PLC interactions have been verified in pea and wheat, these findings indicate that G proteins and/or receptor kinases are likely to act as thermosensors (Misra et al., 2007; Khalil et al., 2011; Munnik, 2014). However, whether similar mechanisms occur in the plant kingdom still needs further investigation, and the demonstration of molecular responses provides indispensable clues to explore thermosensors in plants.

Phase separation in protein levels

SG (stress granule) formation is the output of overall repression of translation initiation, and protein LLPS (liquid-liquid phase separation) is required for SG formation; the condensation of RBGD2 (RNA-binding glycine-rich D2) and RBGD4 into heat-induced SGs based on LLPS is necessary for thermotolerance, where RGBD2/4 gain access to interact with additional SG proteins and heat-responsive mRNA, eventually evoking downstream heat responses (Figure 3; Zhu et al., 2022). In addition, ALAB (acetylation lowers binding affinity) proteins undergo LLPS under HS, form SGs and processing bodies, and ensure thermotolerance by directly binding and stabilizing *HSF* mRNAs (Figure 3) (Tong et al., 2022). Taken together, these findings suggest that protein conformation plays essential roles in temperature perception.

Subcellular location-specific changes in protein levels

Subcellular translocation in response to HS, which alters proteins' interaction with other molecular components, is another countermeasure for thermoperception in plants. A representative pattern is the movement toward the nucleus (Zhu et al., 2023). For example, the glycolytic enzyme GAPC (glyceraldehyde-3-phosphate dehydrogenase) in the cytoplasm relocates to the nucleus, where it interacts with and activates the transcription factor NF-YC10 (nuclear factor Y subunit C10), enabling signaling transmission upon exposure to excess heat (Figure 3; Kim et al., 2020). The cytosol-to-nuclear relocation of GAPC is dependent on PA and PLDδ (Kim et al., 2022b). High temperatures induce membrane-to-nuclear translocation of OsNTL3 (a NAM, ATAF1/2, CUC1/2 [NAC] transcription factor) and elicit downstream HS responses (Figure 3; Liu et al., 2020).

In addition to nuclear translocation, communication between cellular organelles is another approach by which plants sense increases in temperature. TT3.1, localized at the plasma membrane, can respond to and decode extracellular HS signals by translocating into intracellular endosomes (Figure 3) (Zhang et al., 2022a). These results suggest that protein translocation is of great significance in thermosensing.

WELL-ESTABLISHED THERMAL RESPONSES IN *ARABIDOPSIS*

Transcriptional regulation: The central roles of HSF and HSP genes

Activation of *HSR* gene expression lies at the center of heat responses; it plays multifaceted roles in downstream pathways and confers adaptations to HS in plants. In *Arabidopsis*, HsfA1 proteins, members of the HSF family, act as master activators of HSR genes and positively regulate thermotolerance. Gene expression levels and protein activity function in a heat-induced manner (Ohama et al., 2017; Ding et al., 2020). To prevent HsfA1 proteins from being activated under normal temperature conditions, HSP70 and HSP90 form a complex with HsfA1s, blocking

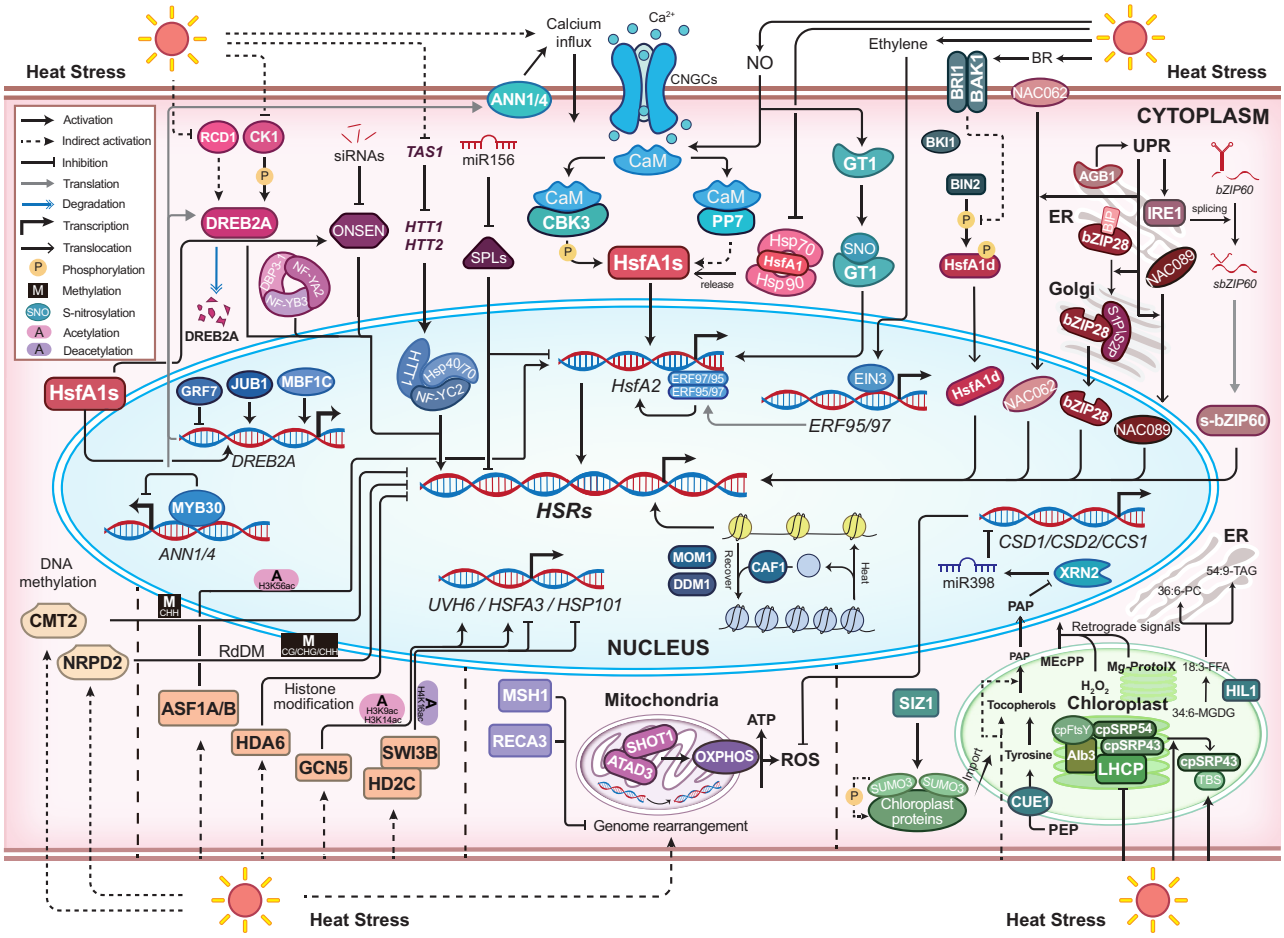


Figure 4. Molecular responses to HS in *Arabidopsis*.

HSF–HSP function predominantly in HSRs, where HS activates HsfA1s by removing the inhibition of Hsp70 and Hsp90. Under HS, HsfA2 and DREB2A are activated by HsfA1, which activates expression of HSR genes. Under normal conditions, the amount of DREB2A is tightly controlled by RCD1 and CK1 at the protein level and by GRF7 at the transcriptional level. Under HS, HsfA1s, JUB1, and MBF1c induce the expression of *DREB2A*, which activates HSR gene expression in a trimer (NF-YB3, NF-YA2, and DPB3-1)-dependent manner. Signaling molecules also play significant roles in the HSR. PP7 and CBK3 affect HsfA1 activities through their CaM-binding domains. GT-1 enhances the expression of *HsfA2* through its *S*-nitrosylation by S-nitrosoglutathione (GSNO);. Ethylene activates HSR genes via the EIN3–ERF95/ERF97–HsfA2 transcriptional cascades. BR switches HsfA1d to its active state by removing BIN2. Heat-induced miR156 and miR398 activate the HSR by post-transcriptionally downregulating HSR inhibitor SPLs and antioxidant enzymes (CSD1, CSD2, and CCS), respectively. siRNAs repress the heat-triggered induction of ONSEN. HS eliminates HTT1/2 inhibition by TAS1, and the increased HTT1 functions as a cofactor to activate HSR genes together with NF-YC2 and Hsp40/70. CMT2 downregulates HSR gene expression by methylation. NRPD2 and HDA6 inhibit the HSR and positively regulate thermotolerance via the RdDM pathway. GCN5 increases *HSF3A* and *UVH6* transcription via the modified histones H3K9ac and H3K14ac (acetylation). HD2C interacts with SWI3B to downregulate *HSFA3* and *HSP101* transcription by H4K16 deacetylation. ASF1A/1B positively regulate *HsfA2* transcription by H3K56ac. CAF-1, DDM1, and MOM1 are required for the reloading of nucleosomes during recovery from HS. HS triggers the unfolded protein response (UPR) in the ER, where ER membrane-localized *bZIP60* mRNA is spliced by IRE1 to generate nucleus-localized sbZIP60, and ER-localized bZIP28 enters the nucleus via S1P/S2P processing in the Golgi apparatus. AGB1 is associated with plant resistance to UPRs. NAC062 and NAC089 exhibit plasma membrane-to-nucleus and ER-to-nucleus translocation, respectively. The rearrangement of the mitochondrial genome is repressed by MSH1 and RECA3. OXPHOS is controlled by SHOT1 and ATAD3 in mitochondria. In chloroplasts, protein import relies on SIZ1 and SUMO3. Chloroplast-localized HIL1 determines lipid metabolism in the ER. Heat disassociates the cpSRP54–cpSRP43 complex, releasing free cpSRP43 to target tetrapyrrole biosynthesis (TBS). Chloroplast-to-nucleus retrograde signaling molecules also play important roles in thermal signal transduction, including H_2O_2, Mg-ProtoIX, PEP, MEcPP, and thylakoid-derived Ca^{2+}, which are encoded by downstream receptors. 3′-Phosphoadenosine 5′-phosphate (PAP) inhibits nuclear exoribonucleases (XRNs) and guarantees that miR398 confers thermotolerance.

their activity and sequestering them from the nucleus, while HS activates HsfA1s by loosening the complex and releasing active HsfA1s (Figure 4; Ding et al., 2020). HsfA1 proteins are also subjected to post-translational modification (PTM), and phosphorylation and SUMOylation can positively or negatively regulate HsfA1 activity, respectively (Ohama et al., 2017; Rytz et al., 2018; Ding et al., 2020). In addition, a more comprehensive mechanism of HSF1 has been reported in mammals; for example, phosphorylated HSF1 can form small nuclear condensates via LLPS to promote downstream transcription, and this process is reversible, with HS responses switched off when HSF1 condensates are dispersed (Zhang et al., 2022). Whether a similar mechanism exists in plants remains an enigma, and further experimental studies are needed to reach a conclusion.

The best known of the downstream HsfA1s, DREB2A (DEHYDRATION-RESPONSIVE ELEMENT BINDING 2A), is a primary regulator of drought and heat responses, and the mRNA and protein levels are precisely regulated in response to HS. GRF7 (GROWTH REGULATING FACTOR 7) represses the actions of DREB2A under normal conditions. Under HS, HsfA1s, JUB1 (JUNGBRUNNEN 1), and MBF1c (MULTIPROTEIN BRIDGING FACTOR 1C) act to increase the mRNA levels of *DREB2A*, while RCD1 (RADICAL-INDUCED CELL DEATH 1), a destabilizer of DREB2A, is degraded. The NRD (NEGATIVE REGULATORY DOMAIN) of DREB2A is phosphorylated by CK1 (CASEIN KINASE 1), which determines its degradation, and HS reduces the phosphorylation of NRD, leading to increased DREB2A abundance (Ding et al., 2020). The activation of DREB2A for the downstream HSR cannot occur without the assistance of a trimer composed of NF-YB3 (NUCLEAR FACTOR Y), NF-YA2, and DPB3-1 (DNA POLYMERASE II SUBUNIT B3-1) (Figure 4; Ohama et al., 2017; Ding et al., 2020). Besides, transcriptional signal transduction is not solely dependent on HSF/HSP. The circadian clock proteins REVEILLE4 (RVE4) and RVE8 have been reported to initiate HS-inducible gene expression independent of HSF (Li et al., 2019).

Signal molecules: Ca^{2+}, ROS, nitric oxide, and phytohormones

In response to heat, Ca^{2+} acts as a second messenger that binds directly to CaM (calmodulin) and is subsequently decoded by CaM. Phosphorylation of HsfA1 proteins by CBK3 (CaM-binding protein kinase 3) determines their binding affinity to downstream target promoters (Figure 4; Liu et al., 2008). Another interactor of HsfA1s is a CaM-binding phosphatase, PP7; however, it is unclear whether PP7 directly dephosphorylates HsfA1 (Figure 4; Liu et al., 2007). In addition to Ca^{2+}, ROS (e.g., hydrogen peroxide [H_2O_2], superoxide [$O2^-$], singlet oxygen [$^1O_2$], and hydroxyl radicals OH^-]) can be classified as signaling and metabolic types. Signaling ROS is generated by nicotinamide adenine dinucleotide phosphate ROS is produced by intracellular organelles (e.g., chloroplasts, mitochondria, and peroxisomes). When the accumulated ROS elicit cell damage, antioxidative and ROS scavenging systems are activated to maintain cellular redox homeostasis and HS acclimation (Choudhury et al., 2017). Extracellular H_2O_2 serves as a signal molecule to directly activate relevant sensors; e.g., the leucine-rich-repeat receptor kinase HPCA1 senses H_2O_2 based on cysteine residues and eventually contributes to Ca^{2+} elevation (Wu et al., 2020). However, it is unknown whether these sensors participate in HS responses. In addition, H_2O_2 may also be upstream of HsfA1 activation, which, in turn, initiates transcription of heat shock-related genes (Volkov et al., 2006; Guan et al., 2013). Apart from ROS, reactive nitrogen [N] species also trigger *S*-nitrosylation of the transcription activator GT-1 via S-nitrosoglutathione to enhance its binding to the *HsfA2* promoter for HS responses in *Arabidopsis* (Figure 4; He et al., 2022). Moreover, plant hormones have multiple functions in the response to HS. For example, ethylene activates two transcription factors, ERF95 (ETHYLENE RESPONSE FACTOR 95) and ERF97, via the key transcription factor EIN3 (ETHYLENE INSENSITIVE3) to enhance *HsfA2* expression and confer basal thermotolerance (Huang et al., 2021). BRs (brassinosteroids) also positively regulate thermotolerance in *Arabidopsis*; in the absence of BRs, BIN2 (BRASSINOSTEROID INSENSITIVE2) predominantly interacts with and phosphorylates HsfA1d, diminishing the nuclear localization and DNA-binding affinity of HsfA1d. However, when BRs are added, the activity of BIN2 is largely inhibited, rescuing nuclear translocation and the DNA-binding ability of HsfA1d, and subsequently activating HSR (Figure 4; Luo et al., 2022).

ncRNAs

Plant ncRNAs should not be ignored in epigenetic HSRs; they work upstream of the main transcription factors or genes. MicroRNAs (miRNAs), a class of ncRNAs, target mRNAs to promote their degradation or block their translation. Under normal conditions, HSR genes are strictly controlled by the master transcription factor SPL (SQUAMOSA PROMOTER BINDING PROTEIN-LIKE) proteins, and after HS, *miR156* isoforms are highly induced. Expression of *SPL* genes is post-transcriptionally downregulated by *miR156*, which contributes to activating the expression of HSR genes and facilitating HS memory (Figure 4; Stief et al., 2014). miR398 expression is also rapidly induced by heat, which decreases *CSD1/2* (copper/zinc superoxide dismutase [SOD] 1/2) and *CCS* (copper chaperone for SOD) expression and ultimately activates HSR and ROS accumulation (Figure 4; Guan et al., 2013). Given that HsfA1b and HsfA7b are required for heat-induced miR398 accumulation, a positive feedback loop has been established among HSF, *miR398*, and *CSD1/2/CCS* (Guan et al., 2013).

siRNAs (small interfering RNAs) are endogenous RNA molecules 21–24 nt in length that, in addition to regulating gene expression, also participate in modulating HSRs. Indeed, HS induces the activation of the *copia-like* retrotransposon ONSEN (japanese 'hot spring'), which is a target of HsfA1 and HsfA2, and triggers a burst of transposition in the plant genome. The siRNA pathway is indispensable for restricting ONSEN transposition activity (Figure 4; Ito et al., 2011). *Trans*-acting siRNAs (ta-siRNAs) are a special type of siRNAs that are processed from non-coding ta-siRNA precursors. TAS1 (ta-siRNA precursor 1)-derived siRNA targets HTT1 (HEAT-INDUCED TAS1 TARGET1) and HTT2 and is involved in thermotolerance regulation (Li et al., 2014). Heat-inducible *HTT* gene expression is negatively regulated by TAS1 but positively regulated by HsfA1a, contributing to the activation of HSR, where HTTs function as cofactors of the Hsp70–NF–YC2 complex to elicit heat responses (Li et al., 2014; Figure 4).

Epigenetic modification

Epigenetic regulation of the heat response includes DNA methylation, histone modification, and chromatin remodeling. For DNA methylation, CMT2 (chromomethylase 2)-associated CHH (C= cytosine, H = any base but guanine) methylation levels are negatively correlated with thermotolerance in *Arabidopsis* (Shen et al., 2014). In contrast, NRPD2 (the second-largest subunit of RNA Pol IV and V) is a positive regulator of thermotolerance (Popova et al., 2013). Furthermore, the molecular mechanisms that determine how remnant transposable elements controlled by RdDM (RNA-directed DNA methylation) result in expression of adjacent protein-coding genes have been well established in *Arabidopsis*, indicating that DNA methylation plays a vital role in thermotolerance regulation (Figure 4; Popova et al., 2013; Zhao et al., 2021a).

Histone modification is achieved by numerous epigenetic regulators, such as acetyltransferases, methyltransferases,

deacetylases, and demethylases, to regulate gene expression under HS conditions. The histone acetyltransferase GCN5 (general control non-depressible 5) positively regulates *HSFA3* and *UVH6* (ULTRAVIOLET HYPERSENSITIVE6) expression through H3K9/K14 acetylation in their promoters and confers thermotolerance in *Arabidopsis* (Figure 4; Hu et al., 2015). Under HS, the transcriptional repressor *Arabidopsis* deacetylase HD2C interacts with SWI3B (a subunit of the ATPases BRAHMA-containing SWITCH/SUCROSE NONFERMENTING [SWI/SNF] chromatin-remodeling complex) to downregulate the expression levels of *HsfA3* and *Hsp101* and other HSR genes by H4K16 deacetylation (Figure 4; Buszewicz et al., 2016). HDA6 (histone deacetylase 6), an RPD3-type deacetylase, is associated with repression of gene expression by RdDM and positively regulates thermotolerance (Figure 4; Popova et al., 2013; Zhao et al., 2021a). Upon HS, another histone deacetylase, HDA9, relocates from the cytosol to the nucleus to positively modulate heat shock signal transduction in a phosphatase PP2AB′β- and nucleoporin HOS1-dependent manner (Niu et al., 2022). ASF1A (ANTI-SILENCING FUNCTION 1) and ASF1B serve as histone chaperones and positively function upstream of HsfA2 and HSPs via H3K56 acetylation in response to HS (Figure 4; Weng et al., 2014).

For chromatin remodeling, a high proportion of H3K9 dimethylation and a low proportion of H3K4 tri-methylation are required in plants to prevent HSR from activation under normal conditions. HS decondenses heterochromatin by removing nucleosomes before transcriptional activation of HSR gene expression (Pecinka et al., 2010; Iwasaki and Paszkowski, 2014). During recovery from HS, nucleosomes are reloaded and chromatin is re-silenced, depending on the H3–H4 histone chaperone CAF-1 (CHROMATIN ASSEMBLY FACTOR-1), DDM1 (decrease in DNA methylation1), and MOM1 (Morpheus' molecule 1), where DDM1 and MOM1 are closely associated with stress memory (Figure 4; Pecinka et al., 2010; Iwasaki and Paszkowski, 2014).

UPRs in the ER

Heat-triggered misfolded protein accumulation causes the UPR in the ER, which is regulated by membrane-localized bZIP (basic leucine zipper) and other NAC-type transcription factors (Ding et al., 2020). For example, bZIP60 and bZIP28 can induce the expressions of ER stress-responsive genes under ER stress (Iwata and Koizumi, 2005). IRE1 (INOSITOL-REQUIRING ENZYME 1), which is localized in ER membranes, splices *bZIP60* mRNA to produce sbZIP60 (spliced bZIP60), which is capable of translocating into the nucleus to activate transcription of its target genes; i.e., *BiP* (*BINDING PROTEIN*) genes (Figure 4; Deng et al., 2011; Ding et al., 2020). BiP, an ER-localized chaperone, interacts with and maintains bZIP28 in the ER under normal conditions. However, under ER stress, the lumen-facing C terminus domain of bZIP28 determines its movement from the ER toward the Golgi apparatus and subsequent proteolytic activation (Sun et al., 2013). In the Golgi apparatus, bZIP28 is processed by S1P (site 1 protease) and S2P, and the processed bZIP28 acquires nuclear localization to activate HSR (Figure 4; Ding et al., 2020). Plant sensitivity to UPRs is also positively regulated by the heterotrimeric G protein β subunit AGB1 in *Arabidopsis* (Figure 4; Wang et al., 2007). In response to ER stress, the plasma membrane-localized transcription factor NAC062 (also known as ANAC062/NTL6) relocates to the nucleus, while the transcription factor NAC089 translocates from the ER membrane to the nucleus and subsequently activates the UPR. Interestingly, *NAC089* expression is regulated by both bZIP28 and bZIP60, while NAC062 is under the direct control of bZIP60 (Figure 4; Ruberti et al., 2015).

Molecular regulation in semiautonomous organelles

HS elicits genome rearrangement in semiautonomous organelles, such as mitochondria and chloroplasts. MutS homolog 1 (MSH1) and RECA3 repress this process, and the *msh1 reca3* double mutant leads to extensive rearrangement of the mitochondrial genome, which exhibits enhanced thermotolerance in *Arabidopsis* (Figure 4; Shedge et al., 2010). OXPHOS (oxidative phosphorylation) is an essential metabolic pathway that generates ATP and ROS in mitochondria. SHOT1, a mitochondrial transcription termination factor-related protein, functions dominantly in the normal accumulation of OXPHOS complexes I and IV and interacts with ATAD3 (ATPase family AAA domain-containing protein 3) to sustain mitochondrial nucleoid organization, and plants with mutations in both genes exhibit enhanced thermotolerance in *Arabidopsis* (Figure 4; Kim et al., 2021).

Chloroplasts are vulnerable to HS, which disrupts the photosynthetic system, and plastid gene expression is deregulated, eventually leading to chloroplast malfunction. The import of nuclear-encoded proteins into chloroplasts is indispensable for HS tolerance in plants. The SUMO ligase AtSIZ1 directly SUMOylates these chloroplast preproteins, and SUMO3 conjugations potentially lead to phosphorylation of the preproteins (Figure 4; Zheng et al., 2022). These two processes ensure adequate protein import into chloroplasts and confer thermotolerance in *Arabidopsis* (Zheng et al., 2022). HIL1 (HEAT INDUCIBLE LIPASE 1) functions in chloroplasts to release 18:3-free fatty acids from 34:6-monogalactosyl diacylglycerols, and free 18:3-free fatty acids is subsequently converted into 36:6-phosphatidylcholines and 54:9-triacylglycerols in the ER, remodeling the lipid composition upon HS (Figure 4; Higashi et al., 2018). The molecular chaperone cpSRP43 (chloroplast signal recognition particle 43) modulates the post-translational regulation of light-harvesting chlorophyll-binding proteins in the thylakoid membrane and also plays a vital role in the protection of tetrapyrrole biosynthesis (TBS) (Ji et al., 2021). cpSRP43 is capable of preventing TBS proteins from aggregation and stabilizes them upon HS. Under normal conditions, cpSRP43 is also responsible for light-harvesting chlorophyll a/b protein, a subunit of photosystem II (PSII), integration into the thylakoid membrane, by interacting with cpSRP54. However, HS loosens the cpSRP54-cpSRP43 interaction, releasing free cpSRP43 that preferentially binds to TBS proteins to protect them from stress-induced aggregation (Figure 4; Ji et al., 2021).

In addition, over-accumulation of metabolites, including ROS, tetrapyrroles, heme, MEcPP (methyl-D-erythritol 2,4-cyclodiphosphate), and even Ca^{2+}, was detected in chloroplasts (Sun and Guo, 2016; Wu and Bock, 2021; Richter et al., 2023). These molecules function in plastid-to-nucleus signal transduction and are regarded as retrograde signals. HS induces a spectrum of cellular responses. For example, ROS bursts and redox imbalance trigger downstream responses via the nuclear transcription factors golden-like transcription factor, HY5 (ELONGATED

HYPOCOTYL 5), or ABI4 (abscisic acid [ABA] INSENSITIVE 4) (Sun and Guo, 2016). Stromal Ca^{2+} levels are elevated by the thylakoid membrane Ca^{2+} sensor and activate the downstream signaling transduction process in the nucleus via CaM/Ca^{2+}-binding protein kinase and transcription factors (Lenzoni and Knight, 2019). Mg-protoporphyrin IX, a canonical tetrapyrrole, directly binds to HSP90 (Kindgren et al., 2012). HSP90.1 interacts with peptidyl prolyl isomerase ROF1 (rotamase FKBP1) to form a complex that subsequently moves into the nucleus in dependent on HsfA2 and guarantees sHSP (small HSP) accumulation (Meiri and Breiman, 2009). MecPP induces the expression of *IRE1a* and *bZIP60*, which are involved in UPRs via CAMTA3 (CaM-binding transcription activator 3) (Figure 4; Benn et al., 2016; Sun and Guo, 2016; Hayes et al., 2021).

The extra-plastid phosphoenolpyruvate (PEP) is imported into the chloroplast by a PEP translocator, CUE1 (chlorophyll a/b binding protein [CAB] gene underexpressed 1) to produce tocopherols (vitamin E) via the aromatic amino acid tyrosine (Fang et al., 2019). Tocopherols are indispensable for the accumulation of 3′-phosphoadenosine 5′-phosphate, which acts as a retrograde signal to inhibit the activity of nuclear exoribonucleases, which facilitates primary miRNA stabilization and mature miRNA production (e.g., miR398), thereby conferring acquired thermotolerance in *Arabidopsis* (Figure 4) (Fang et al., 2019). These results provide evidences that retrograde signaling derived from the chloroplast also plays indispensable roles in molecular responses to HS.

MOLECULAR REGULATION AND GENETIC CONTROL OF HSRs IN CROPS

Crop yields continue to decrease under the current climate of global warming (Zhao et al., 2017). Therefore, understanding the molecular responses of crops to HS is crucial for future agricultural advancement. Rice is grown worldwide, from temperate to tropical regions, and some subspecies that originated in Africa or other tropical regions, such as African cultivated rice (*Oryza glaberrima*) and tropical *japonica*, have undergone geographical adaptations and acquired numerous natural alleles that confer resilience to different environmental niches. The network of HS responses in rice, maize, and tomato is elaborated in this section.

Calcium elevation initiates heat signal transduction

Changes in membrane status are always regarded as the initial responses of plants to HS, followed by activation of receptors, which subsequently contribute to increases in cytosolic Ca^{2+} and protein translocation, among other effects. OsCNGC14 (cyclic nucleotide-gated ion channel) and OsCNGC16 have been shown to be essential for heat-triggered Ca^{2+} influx in rice (Figure 5; Cui et al., 2020). However, it was not until the identification of TT2 (Thermotolerance 2) in rice that the process of how the heat-triggered Ca^{2+} elevation is decoded into a downstream response became unambiguous (Kan et al., 2022).

Previous studies reported that the roles of G proteins are widely recognized in development and production improvement in rice; however, the functions of G proteins in thermotolerance remain unclear. *TT2* (*GS3*) encodes a G protein gamma subunit, and the natural allele $TT2^{HPS32}$ from tropical *japonica* rice confers thermotolerance in rice (Kan et al., 2022). A premature stop codon is present in the $TT2^{HPS32}$ locus, suggesting that TT2 functions as a negative regulator of thermotolerance (Kan et al., 2022). TT2 is required for heat-triggered increases in the cytosolic Ca^{2+} level, which is subsequently decoded by SCT1-CaM. The SCT1-CaM interaction is closely linked with cytosolic Ca^{2+} concentration, and with normal TT2 function, higher Ca^{2+} levels promote CaM binding to SCT1 (sensing Ca^{2+} transcription factor 1), whose activity is suppressed by CaM, reducing the target gene expression (i.e., *OsWR2*, *wax synthesis regulatory 2*) and thus exhibiting wax reduction and thermosensitivity under HS (Figure 5; Kan et al., 2022). The $TT2^{HPS32}$ null allele abolishes heat-triggered increases in Ca^{2+} levels, thereby impairing the CaM-SCT1 interaction, and the suppression of SCT1 is released. This contributes to the maintenance of normal *OsWR2* levels, which finally leads to normal wax content and enhanced thermotolerance in rice (Kan et al., 2022). TT2 is a rare negative regulator of thermotolerance and therefore could be highly valuable when used in agricultural production.

Plasma membrane-localized proteins translocate into other cell components to transduce heat signals

The plasma membrane acts as the front line in sensing extracellular heat signals; hence, plasma membrane-localized protein translocation plays identically indispensable roles in bridging the communications among different organelles. OsANN1 accumulates in the cytosol from the cell periphery in response to a 10-min heat treatment (Figure 5; Qiao et al., 2015). OsNTL3 shifts from the plasma membrane to the nucleus in response to HS and activates transcription (Liu et al., 2020). ZmCDPK7 (calcium-dependent protein kinase) translocates from the plasma membrane to the cytosol in response to HS and phosphorylates sHSP17.4 (small HSP 17.4) and RBOHB and upregulates their expression levels (Zhao et al., 2021b). Expression of the genes for these three proteins is heat inducible and dependent on abscisic acid. RBOHB also exerts a feedback role upstream of sHSP17.4 and ZmCDPK7 (Figure 5; Zhao et al., 2021b).

Chloroplasts, as energy suppliers, are highly vulnerable to the effects of HS. Therefore, communication between the plasma membrane and chloroplast plays an important role in orchestrating plant thermotolerance. The *TT3* locus consists of two genes; *TT3.1* encodes a transmembrane E3 ligase, and *TT3.2* encodes a chloroplast precursor protein, which are positive and negative regulators of thermotolerance, respectively (Zhang et al., 2022a). TT3.1 is translocated from the plasma membrane to the endosomes in a heat-dependent manner, while TT3.2 is subsequently ubiquitinated by TT3.1 and removed via the vacuolar degradation pathway, indicating that TT3.1 functions as a putative thermosensor, bridging extracellular heat signals and intracellular chloroplast responses (Figure 5; Zhang et al., 2022a).

Over-accumulation of mature TT3.2 in chloroplasts is detrimental to the thylakoids; therefore, ubiquitination and degradation of TT3.2 by TT3.1 is required in rice to maintain normal chloroplast function under HS (Zhang et al., 2022a). The *TT3.1–TT3.2* genetic module located in a single locus provides a valuable genetic resource for breeding thermotolerant crops, and a comprehensive explanation

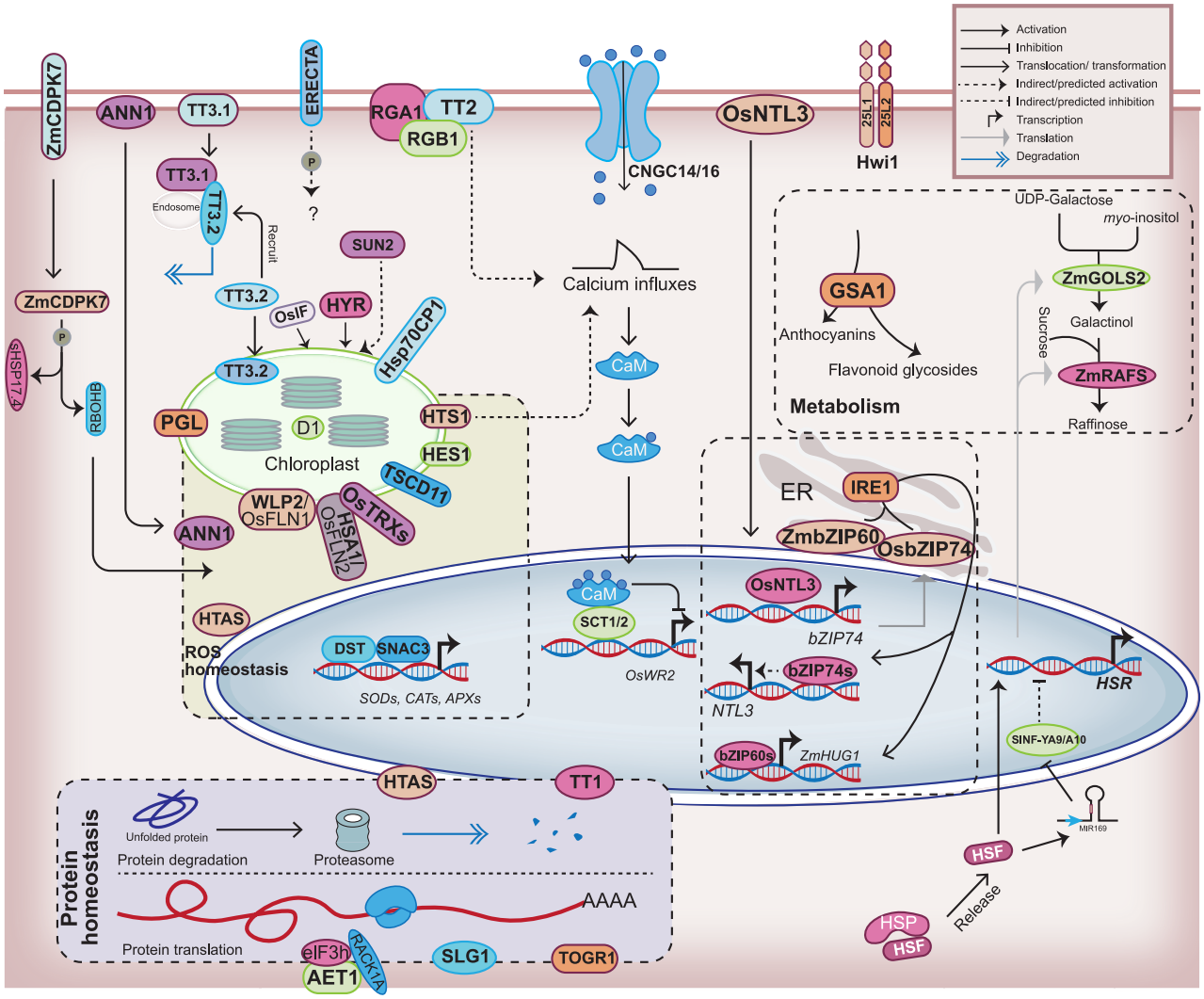


Figure 5. Molecular responses to HS in crops.

In rice, CNGC14/16 facilitate heat-triggered Ca^{2+} influx. TT2 also ensures the heat-inducible increase in cytosolic Ca^{2+} concentration, which is subsequently decoded by CaM-SCT1/2 interactions, resulting in the inhibition of *OsWR2* transcription in rice. ERECTA and Hwi1 work as potential receptor-like kinases and function upstream of the heat responses. In response to HS, TT3.1 translocates from the plasma membrane to the endosomes, where TT3.2 is recruited and degraded by TT3.1 via ubiquitination, preventing the accumulation of mature TT3.2 in chloroplasts in rice. OsNTL3 performs heat-triggered membrane-to-nucleus translocation and transcriptionally activates downstream genes, including *bZIP74*. In rice and maize, ER-localized OsbZIP74 and ZmbZIP60 are spliced by IRE1, and these two transcription factors participate in the transcriptional activation of *OsNTL3* and *ZmHUG1*, respectively. The PSII component D1, along with OsIF, PGL, and HES1, which are required for maintenance of chloroplast ultrastructure, is sensitive to HS in rice. Hsp70CP1 and TSCD11 regulate chloroplast development under HS in rice. HYR and NSUN2 guarantee photosynthetic gene transcripts in rice. HTS1 controls lipid metabolic flux under HS in rice. PGL, HTS1, HES1, and TSCD11, as well as the OsTRXz–OsFLN2–OsFLN1 complex, maintain the redox balance in the chloroplasts. ANN1, DST, SNAC3, and HTAS are involved in ROS scavenging in response to heat. TT1 and HTAS function in the elimination of unfolded proteins, and RACK1A–eIF3h–AET1, SLG1, and TOGR1 guarantee protein translation under HS conditions in rice. The HSP–HSF complex is loosened in response to HS, and HSFs are subsequently released, and the inhibition of NF-YA9/A10 for HSR is removed by Hsfs-mediated accumulation of miR169s in tomato. In metabolism, *GOLS2* and *RAFS* are transcriptionally activated by HSF in maize, and GSA1 positively regulates thermotolerance by modulating flavonoid glycoside and anthocyanin fluxes in rice.

of the TT3.1-TT3.2 regulatory module increases our understanding of the molecular mechanisms that underly crop heat responses (Zhang et al., 2022a; Wang and Cao, 2022).

Chloroplasts as hubs of energy supplies and ROS metabolism upon HS

Chloroplasts bear the brunt of HS; thus, preventing damage to chloroplasts is the first priority for plants to survive HS. An adequate supply of PSII proteins, such as the D1 subunit, expressed from genes driven by artificial promoters, increases thermotolerance and grain yield in rice (Figure 5; Chen et al., 2020). Preventing import of the precursor protein TT3.2 into chloroplasts protects thylakoids from HS-induced damage (Figure 5; Zhang et al., 2022a). OsIF (intermediate filament), PGL (pale green leaf), and HES1 (enhanced lesion spots 1) have been reported to maintain chloroplast ultrastructure, and plants

carrying mutations in the genes for these proteins display thermosensitivity to HS (Yang et al., 2016; Soda et al., 2018; Xia et al., 2022). Moreover, chloroplast differentiation from proplastids under elevated temperatures is dependent on OsHsp70CP1, and functional TSCD11 (temperature-sensitive chlorophyll-deficient 11) is required for early chloroplast development under HS (Figure 5; Kim and An, 2013; Fang et al., 2020). HYR (higher yield rice), a transcription factor that participates in photosynthetic carbon metabolism, has been shown to promote photosynthesis under HS conditions (Ambavaram et al., 2014). Heat-induced 5-methylcytosine modification of mRNA involved in photosynthesis, is dependent on OsNSUN2, an RNA 5-methylcytosine methyltransferase (Figure 5; Tang et al., 2020). Thylakoid membrane-localized HTS1 (high temperature sensitive 1) guarantees fatty acid synthesis and lipid metabolism in plants exposed to HS (Figure 5; Chen et al., 2021). Upon exposure to HS, SlCDJ2, a chloroplast-targeted DnaJ protein, keeps low levels of proteolytic enzyme activity, which reduces the rate of ribulose 1,5-bisphosphate carboxylase/oxygenase (RuBisCO) degradation to maintain CO_2 assimilation in tomato (Wang et al., 2015).

In addition to maintaining chloroplast function, eliminating oxidative damage is equally important for chloroplasts to deal with heat-induced damage. The TRX-FLN (thioredoxin-fructokinase-like) regulatory pathway, composed of the pfkB-type carbohydrate kinase WLP2 (white leaf and panicle 2), OsFLN2 (its paralog, fructokinase-like 2), and OsTRXz (thioredoxin), maintains redox homeostasis under HS in chloroplasts (Figure 5; Lv et al., 2017). Loss of function in the above-mentioned chloroplast-associated proteins (including PGL, HTS1, HES1, and TSCD11) results in excessive ROS accumulation and thermosensitivity in rice (Figure 5; Yang et al., 2016; Fang et al., 2020; Chen et al., 2021; Xia et al., 2022). In addition to chloroplasts, proteins in other cellular components also perform ROS scavenging to maintain redox homeostasis. Cytoplasm-localized OsANN1 boosts the activities of SOD and catalase (Qiao et al., 2015). The nucleus-localized transcription factors DST (drought and salt tolerance) and OsSNAC3 (stress-responsive NAC) have both been shown to regulate the expression of genes encoding SOD, catalase, and ascorbate peroxidase under HS (Huang et al., 2009; Fang et al., 2015; Zhang et al., 2019). A cytoplasm- and nucleus-localized ubiquitin (Ub) ligase, OsHTAS (heat tolerance at seedling stage), modulates H_2O_2 accumulation and confers thermotolerance in rice (Figure 5; Liu et al., 2016). To promote HSR gene expression and balance the level of programmed cell death, tomato seedlings increase the content of nicotinamide adenine dinucleotide phosphate oxidase-mediated H_2O_2 and control the quantity of chloroplast-derived H_2O_2, eventually improving their ability of maintenance in acquired thermotolerance (Sun et al., 2018).

HSF–HSP-mediated transcription activation in the nucleus

Multiple isoforms of HSP and Hsf (HS transcription factors) have been identified that form complexes and function downstream in determining transcriptional levels. In *Arabidopsis*, a systematic network centered on HSF–HSP has been well established (Figure 4; Ding et al., 2020). Similar regulatory networks centered on HSF–HSP have been obtained in crops, except for a few supplements. In tomato, Hsf proteins (e.g., HsfA1a, HsfB1, HsfA7s, and HsfA6b) are capable of binding to the promoter of *MIR169* and cause accumulation of miR169s (Hahn et al., 2011). The elevated miR169 levels subsequently decrease the levels of NF-YA9/A10, thus eliminating the suppression of NF-YA9/A10 on HSR genes such as *HsfA2*, *HsfA3*, and *HsfA7* (Hahn et al., 2011; Rao et al., 2022). In addition, a CREB binding protein-like protein, HAC1, participates in HSR activation and HSG maintenance by forming a ternary complex with HSFs in tomato (Bharti et al., 2004; Hahn et al., 2011; Rao et al., 2022). OsHSP1 positively regulates thermotolerance in *Arabidopsis*, and a positive feedback loop between OsHSP101 and HSA32 (HS-associated 32-kD protein) determines long-term acquired thermotolerance in rice (Lin et al., 2014; Moon et al., 2014). In maize, ZmHSFA2 undergoes PTM and forms active homotrimers in a heat-dependent manner (Gu et al., 2019). *ZmHSBP2* (*HEAT SHOCK BINDING PROTEIN 2*) expression is activated by ZmHSFA2. ZmHSBP2 interacts with the trimers to destroy the promoter binding to the heat shock element, attenuating the HSRs and constituting a negative feedback-type regulation (Gu et al., 2019).

Regulators of protein degradation and translation maintain protein homeostasis under HS

Disruption of protein homeostasis (i.e., the over-accumulation of cytotoxic proteins and abnormal protein translation) can lead to cell damage and even cell death. Therefore, renaturing or eliminating these unfolded proteins is necessary for plants to deal with these situations (Zhang et al., 2019). To our knowledge, TT1 is the first natural allele identified in crop thermotolerance, and it encodes an α2 subunit of the 26S proteasome, a large multisubunit protease complex that targets ubiquitinated proteins for degradation (Figure 5; Li et al., 2015). The $TT1^{CG14}$ allele that originated from African rice is more efficient in eliminating denatured proteins and mitigates heat response fluctuation at the level of Ub-modified proteins (Li et al., 2015). The *TT1*^{CG14}-harboring nearly isogenic line, which exhibits enhanced thermotolerance, shows excellent performance in the heading and grain filling stages under HS (Li et al., 2015). In addition, excellent thermotolerance is also shown in transgenic plants of other species (e.g., *Arabidopsis* and tall fescue) that overexpress *TT1*^{CG14} (Li et al., 2015). OsHTAS can effectively eliminate cytotoxic proteins via the ubiquitination pathway, sharing similar manners with TT1 (Liu et al., 2016). SlCHIP (carboxyl terminus of the hsc70-interacting protein), which encodes a chaperone-dependent Ub E3 ligase, targets misfolded proteins and positively regulates thermotolerance in tomato (Zhang et al., 2021).

Ensuring normal protein translation is equally required for plants to maintain protein homeostasis under HS. The stability of pre-rRNA is essential for protein translation, and the heat-induced DEAD (Asp-Glu-Ala-Asp)-box RNA helicase, TOGR1, plays an essential role in this process and confers thermotolerance when overexpressed in rice (Figure 5; Wang et al., 2016). Sufficient thiolated tRNA levels help plants to survive exposure to HS, where *SLG1* (*Slender Guy 1*), a gene that encodes tRNA 2-thiolation protein 2, positively regulates thermotolerance in rice through the tRNA thiolation pathway (Figure 5; Xu et al., 2020b). The pre-tRNA^{His} modification is vulnerable to HS, and AET1, RACK1A, and eIF3h form a complex to protect this process

and modulate auxin signaling at the translational level under HS (Figure 5; Chen et al., 2019).

Post-translational modification

The alteration of modification in protein levels can also be regarded as a response to HS in plants. In tomato, a well-identified SUMO E3 ligase, SlSIZ1, enhances thermotolerance by mediating the SUMOylation of HsfA1 (Zhang et al., 2018b). SlMPK1 (mitogen-activated protein kinase 1) directly phosphorylates SlSPRH1 (serine-proline-rich protein homolog) to down-regulate the content of antioxidant defense proteins, which negatively controls thermotolerance in tomato (Ding et al., 2018). Some receptor-like proteins, such as ERECTA and Hwi1, localize to the plasma membrane and employ unidentified mechanism to bridge extracellular heat signals and intracellular physiological responses via putative phosphorylation (Figure 5; Chen et al., 2014; Shen et al., 2015).

URPs in the ER under HS

UPRs are activated in the ER under HS. Subsequently, bZIP transcription factor family were processed and were granted the access to the nucleus in plants (Lu et al., 2012; Liu et al., 2020). The regulatory pathway in crops (i.e., rice and maize) shares large similarities with that in *Arabidopsis* (Lu et al., 2012; Liu et al., 2020). The double stem-loop structure in ER-associated *OsbZIP74* (*OsbZIP50*)/*ZmbZIP60* mRNA is spliced out by IRE1, generating nuclear-access short-form bZIP. The heat-inducible chaperone ZmHUG1 (HEAT UP-REGULATED GENE 1) was subsequently activated to prevent the aggregation of PRA1.C1 (MAIZE PRENYLATED RAB ACCEPTOR 1.C1) and stabilize other putative targeted proteins, thereby alleviating ER stress and conferring thermotolerance in maize (Xie et al., 2022). Moreover, in rice, short-form OsbZIP74 participates in the heat-induced upregulation of *OsNTL3* transcription, and the heat-triggered nuclear-relocated OsNTL3 can directly bind to the promoter of *OsbZIP74* and regulate its transcription in response to heat (Figure 5; Liu et al., 2020).

Metabolism flux

GSA1, which encodes uridine diphosphate-glucosyltransferase, is a positive grain size regulator that redirects metabolic flux from lignin biosynthesis into the flavonoid biosynthesis pathway, facilitating the accumulation of flavonoid glycosides and anthocyanins to eliminate the damage from HS (Dong et al., 2020a). HsfB1 also functions in metabolic pathways, including the phenylpropanoid and flavonoid pathways, which link stress responses with physiological development in tomato (Paupière et al., 2020). The Hsf-metabolite crosstalk is important for thermotolerance in maize. Heat induces the expression of *ZmHSFA2*. The active trimers enhance raffinose biosynthesis by transcriptionally activating the expression of the *ZmGOLS2* (*GALACTINOL* SYNTHASE 2) and *ZmRAFS* (*raffinose synthase*) genes to modulate heat tolerance (Gu et al., 2019).

There are still many other HS responses that cannot be appropriately categorized in the abovementioned pathways. For example, in maize, a mitochondrion-localized metalloprotease, NDL1 (needle 1), is required in thermotolerant reproductive organogenesis via control of auxin homeostasis (Liu et al., 2019). In addition, maintenance of genome stability and changes in metabolic flux are also noteworthy in heat responses. LS1 encodes an RNase H2 subunit and protects cells against DNA damage under conditions of high light and temperature (Qiu et al., 2019). In the future, mining quantitative trait loci/genes and understanding the sophisticated molecular responses under HS to guide the next generation of rice breeding will become an important approach to cope with global warming.

MAINTAINING GRAIN YIELD THROUGH SOURCE–SINK STRATEGIES

Source strategies

Protecting photosynthesis

Photosynthesis is the fundamental process needed for growth and biomass production, but it is also one of the most heat-sensitive physiological processes in plants. Under HS, RuBisCO activity is noticeably suppressed because of heat-unstable RuBisCO activase (RCA) (Wang et al., 2010; Qu et al., 2021). The large isoform RCA_L functions in photosynthetic acclimation under conditions of moderate HS, while the short isoform, RCA_S, mainly contributes to the maintenance of RuBisCO initial activity under normal conditions (Wang et al., 2010). Without the reduction in RuBisCO content, increasing the RCA content can improve the resilience to HS in rice (Figure 6; Qu et al., 2021). Heat-inducible D1 (a key component of PSII), artificially driven by the *AtHSFA2* promoter, continuously rescues the severe loss of D1 protein in PSII and protects chloroplast of transgenic plants against HS. In addition, adequate D1 supplementation leads to outperformance in biomass and grain yield by promoting higher net CO_2 assimilation rates (Chen et al., 2020). The *TT3.1–TT3.2* genetic module provides novel strategies for breeding heat-tolerant crops, especially in terms of photosynthetic acclimation improvement (Figure 6; Zhang et al., 2022a). HYR works as a direct transcriptional activator of photosynthesis genes and is involved in downstream photosynthetic carbon metabolism, ultimately ensuring grain yield in plants exposed to drought and elevated temperatures (Figure 6; Ambavaram et al., 2014). Under HS, overexpressing *OsIF* protects the total grain yield via stabilized chloroplast structure (Figure 6; Soda et al., 2018).

Reducing water loss under HS

HS is always accompanied by drought stress, and increased temperatures accelerate transpiration rates and other non-stomatal water loss. Therefore, fine-tuning the stomatal movements and cuticle contents is indispensable for plants to reduce water loss under HS. Chloroplast-localized OsMDHAR4 (monodehydroascorbate reductase 4) reduces the H_2O_2 content, promotes stomatal opening, and causes water loss upon HS; therefore, it acts as a negative regulator of thermotolerance in rice (Figure 6; Liu et al., 2018). Similarly, the Ub E3 ligase OsHTAS interacts with an ascorbate peroxidase isoform to modulate H_2O_2-induced stomatal closure, which ultimately positively regulates thermotolerance in rice (Figure 6; Liu et al., 2016). The transcription factor DST targets H_2O_2 homeostasis-related genes and negatively regulates H_2O_2-induced stomatal closure (Figure 6; Huang et al., 2009; Zhang et al., 2019). In addition to stomatal transpiration, water also evaporates through the cuticle layers. The TT2-SCT1-*OsWR2* modulatory pathway explains the wax metabolic fluctuation in response to HS, and a

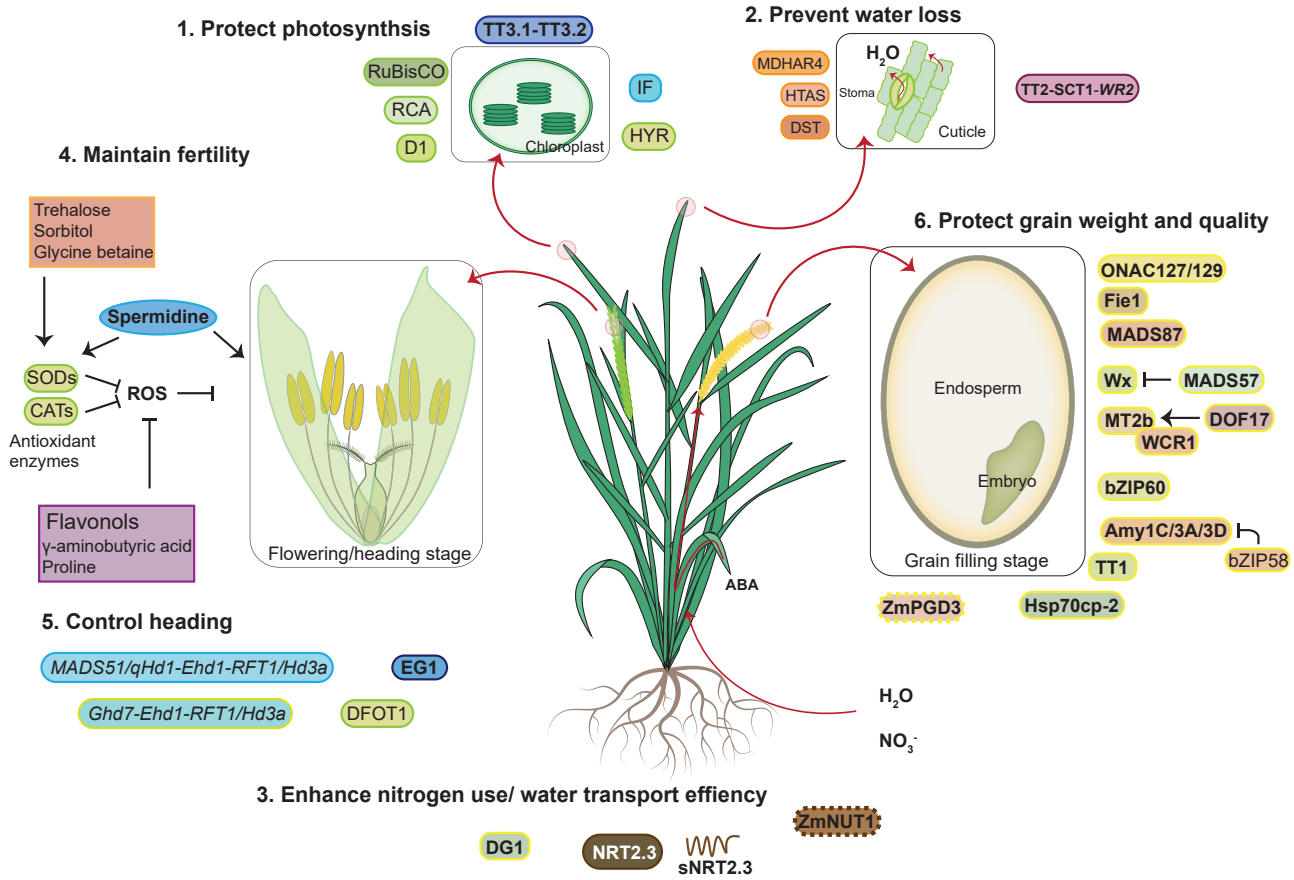


Figure 6. Strategies to improve rice production under HS.

Improving heat acclimation in rice has been proposed from the source–sink perspective. (1) From the source, genetic modulation of RuBisCO, RCA, D1, IF, HYR, and TT3.1–TT3.2 to improve heat tolerance in photosynthesis. (2) Increasing stomatal closure and cuticle deposition to improve water retention via MDHAR4, HTAS, DST, and the TT2-SCT1-WR2 regulatory pathways. (3) Enhancing N use, ABA transport, and water absorption efficiency to increase survival rate via NRT2.3, DG1, and homologs of ZmNUT1. (4) From the sink perspective, exogenous application of carbon and N metabolites or enhancing the endogenous contents to reduce accumulation of ROS in flowering organs to ensure fertilization. (5) Optimization of the heading/flowering stage to avoid heat damage via the *MADS51/qHd1–Ehd1–RFT1/Hd3a* and *Ghd7–Ehd1–RFT1/Hd3a* modulatory pathways or individual genes such as *EG1* and *DFOT1*. (6) Maintaining normal endosperm development, improving starch biosynthesis in the endosperm, and inhibiting amylase to reduce grain chalkiness and improve grain weight and quality via genetic modulation of ONAC127/129, Fie1, MADS87, MADS57-Wx, DOF17–WCR1–MT2b, bZIP60, bZIP58–Amy1C/3A/3D, TT1, Hsp70cp-2, and homologs of ZmPGD3.

natural allele of TT2 abrogates heat-triggered Ca^{2+} influx and blocks Ca^{2+}-dependent CaM-SCT1 interaction, preventing the dramatic reduction in wax content, ultimately conferring thermotolerance in rice (Figure 6; Kan et al., 2022).

Enhancing absorption and transport

Plant growth and production are limited by the availability of N; however, N uptake and utilization efficiency are highly susceptible to heat and drought stress. The nitrate transporter gene *OsNRT2.3* from HTNE-2 (high temperature resistant and nitrogen efficient 2) rice varieties is required to maintain high yields when temperatures at night are high. This results from enhanced translation efficiency of *OsNRT2.3b* and reduced levels of the small RNA (sNRT2.3-1) derived from its 5′ UTR, where sNRT2.3-1 targets *OsNRT2.3a* mRNA depending on ambient temperatures (Figure 6; Zhang et al., 2022b). Especially during the reproductive stage, access to sufficient water transport is important for organ development, fertilization, and grain yield. Therefore, an intact vasculature system plays an indispensable role in long-distance movement, allowing plants to cope with high temperatures and drought conditions. The transcription factor ZmNUT1(necrotic upper tips 1) modulates the reinforcement of cell walls and the maintenance of protoxylem vessels, and its orthologs in rice will be potential targets for the breeding of thermotolerant crops in the future (Figure 6; Dong et al., 2020b). Transport of plant hormones also plays an essential role in production during HS; DG1 (defective grain-filling 1) participates in leaf-derived ABA transport to the caryopsis in a temperature-dependent manner, and *dg1* loss-of-function mutants exhibit poor grain filling and severe grain chalkiness under HS (Figure 6; Qin et al., 2021)

Sink strategies

Guaranteeing normal fertility and heading stage

Compared with the obvious wilting of plant architecture that can occur during vegetative growth, there should be an increased focus on low pollen viability, abnormal fertilization, and poor grain filling in agriculture. Successful fertilization under HS is

determined by two main factors: (1) the contents of harmful metabolites that accumulate in floral organs and (2) the timing of flowering/heading. A reduction of up to 60% in pollen germination, which is usually observed as being due to morpho-anatomical changes that include tapetum degradation, anther indehiscence, and pollen deformity, is correlated with heat-inducible ROS levels under HS (Luria et al., 2019; Santiago and Sharkey, 2019). Some N and carbon metabolites play essential roles in ROS homeostasis, which ensures that fertilization occurs under HS (Santiago and Sharkey, 2019). In rice, exogenous application of spermidine (an important aliphatic polyamine that functions in development, differentiation, stress resistance, senescence, and germination) does not only improve the antioxidant system, preventing ROS accumulation, but also keeps high levels of endogenous spermidine and spermine by decreasing the mRNA levels of *OsSAMDC2*, *OsSPDS1*, and *OsSPDS3*, eventually guaranteeing rice production under HS (Figure 6; Zhou et al., 2020). Flavonols that function as antioxidants also scavenge ROS to maintain redox homeostasis in pollen tube growth and integrity under HS (Figure 6; Muhlemann et al., 2018). In addition to flavonoids, γ-aminobutyric acid (a modulator of calcium-permeable membrane channels) and proline work directly in ROS scavenging, and the four carbon/N metabolites trehalose, sorbitol, glycine betaine, and polyamine stabilize antioxidant enzymes, indirectly quenching ROS levels (Figure 6; Santiago and Sharkey, 2019).

The timing of flowering/heading is another critical feature that determines successful fertilization; therefore, genetically controlling the timing of flowering/heading, either earlier or later, can allow plants to escape from the effects of high temperature and is an excellent strategy for future heat-acclimated crop breeding. A number of genes that control heading have been identified, among which *Hd1* (heading date 1), *Ehd1* (early heading date 1), *Ghd7* (grain number, plant height, and heading date 7), and *Hd3a*/*RFT1* (heading date 3a/rice flowering locus T1) serve as key regulators and form a unique pathway in rice (the Ghd7–Ehd1–Hd3a/RFT1 pathway) (Xue et al., 2008; Zhou et al., 2021). The repression of downstream RFT1 by Ghd7 is alleviated at high temperatures because of a decrease in *Ghd7* mRNA levels in response to heat, which ultimately leads to proper floral induction in rice (Figure 6; Nagalla et al., 2021). Later studies have demonstrated that a natural allele of *qHd1*, which encodes OsMADS51 (a MADS [for MCM1, AG, DEFA and SRF]-box transcription factor), confers thermotolerance at heading and grain filling in rice via the OsMADS51–Ehd1–Hd3a/RFT1 floral pathway (Figure 6; Chen et al., 2018). The expression of floral identity genes such as *OsMADS1*, *OsMADS6*, and *OsG1* is maintained at normal levels under HS by the mitochondrion-localized lipase EG1 (EXTRA GLUME 1), which functions to enhance the robustness of the floral phenotype and act as a buffer to neutralize the disruption caused by high temperature (Zhang et al., 2016). In addition to these canonical floral genes, lodicules-localized DFOT1 (diurnal flower opening time 1) optimizes the timing of flowering by mediating the swelling level of the lodicules, where DFOT1 interacts with PME and promotes its activity (Figure 6; Wang et al., 2022). Taken together, genetic approaches to control flowering can maximize reproductive success by optimally determining the proper timing of phase transition in crops.

Protecting the grain-filling period

HS often contributes to grain-filling attenuation, which includes reduced grain weight and a decline in grain quality. Therefore, maintaining a continuous supply of nutrients from the source to the sink and guaranteeing normal endosperm development are directly related to grain yield under HS conditions. Sugar transport is associated with a heterodimer consisting of ONAC127 (a NAC-type transcriptional factor) and ONAC129 under HS (Figure 6; Ren et al., 2021). The PRC2 (Polycomb repressive complex 2) protein is important for endosperm development in cereals. The transcriptional level of *Fie1*, which encodes a component of the FERTILIZATION INDEPENDENT SEED–PRC2 complex, determines grain width when the night temperatures are high; OsMADS78/79 (MADS box transcription factors) delay cellularization in the endosperm and are negatively regulated by fFERTILIZATION INDEPENDENT SEED-PRC2 (Figure 6; Tonosaki and Kinoshita, 2015; Paul et al., 2020; Dhatt et al., 2021). In addition, OsMADS87 positively regulates seed size but negatively modulates thermotolerance (Figure 6; Chen et al., 2016). Potential allelic variations should be mined based on these genes, which would provide excellent genetic resources for breeding thermotolerant crops.

The endosperm simultaneously determines the weight and quality of cereal grain seeds. Starch is a main component of the endosperm, and it consists of 70%–90% amylopectin with the rest being amylose. The amylose content, which is used to evaluate grain quality, is sensitive to high temperature. The *Wx* (*waxy*) gene is a main regulator of amylose content, and promoting its pre-mRNA splicing efficiency helps plants maintain a stable amylose content and thus improve grain quality under HS (Figure 6; Zhang et al., 2014). OsMADS57 has opposing effects on fertilization and grain filling, and it facilitates spikelet fertility at the early filling stage but destabilizes the endosperm amylose content under HS. Therefore, a specific reduction of OsMADS57 levels in the endosperm could be a valuable approach to improve grain quality under HS (Figure 6; Zhang et al., 2018a).

In plants experiencing HS, protein storage vacuoles are enlarged, but the storage proteins are deficient. Additionally, abnormal development of amyloplasts and elevated levels of amylase expression are both observed in the endosperm, and this contributes to the formation of air spaces during kernel dehydration. The opaque regions in endosperm are therefore regarded as chalkiness, which is an undesirable trait (Wada et al., 2019). OsHsp70cp-2 facilitates protein import into the amyloplast, but plants carrying a mutation in the *OsHsp70cp-2* gene only exhibit grain chalkiness under HS and not under normal conditions (Figure 6; Zhu et al., 2018; Tabassum et al., 2020). The expression level and enzymatic activity of α-amylase are positively correlated with temperature, and over-accumulation of α-amylase (except for Amy3E) causes chalky grains, even in plants not exposed to HS (Figure 6; Nakata et al., 2017). OsbZIP58 functions upstream to positively modulate protein storage in seeds and starch synthesis and to negatively mediate α-amylase. Truncated OsbZIP58β proteins with low transcription activity are more abundant than the full-length OsbZIP58α protein because of heat-induced alternative splicing, which leads to chalky grains (Figure 6; Xu et al., 2020a). In addition, oxidative damage can also lead to chalkiness. *WCR1* (white-core rate 1) encodes an F-box protein that interacts with and stabilizes the metallothionein gene MT2b, reducing ROS

accumulation. An allelic variation in the promoter of WCR1 enhances OsDOF17 binding activity, and the subsequent increase in *WCR1* expression decreases chalkiness and improves grain quality (Figure 6; Wu et al., 2022). In maize, the heat-sensitive 6-phosphogluconate dehydrogenase PGD3 functions in the accumulation of endosperm starch, and when artificially localized to the correct subcellular compartment, this enzyme acquires heat resistance and becomes active, thereby mitigating crop failure caused by HS. Its orthologs in rice can be considered to be potential targets for future investigation (Ribeiro et al., 2020). In conclusion, the revelation of these molecular mechanisms improves our understanding of grain seed development in environments where temperatures fluctuate.

CONCLUDING REMARKS AND FUTURE PERSPECTIVES

The mechanisms underlying the responses of plants to increased temperature have been extensively explored. However, further experimental investigations are needed before a systematic comprehension of these mechanisms can be obtained. The ways in which heat is perceived and then transduced into physiological responses is still unclear. In this review, we divided thermosensing and thermal responses into separate sections to gain a clear understanding of these biological processes and to provide a better summary of these related findings. In fact, we would have to argue that thermal sensing and thermal response processes are interconnected and cooperative, with the former being more mysterious and controversial. Several membrane proteins occupy a primary position in thermosensing, characterized by the acquisition of a combination of extracellular ligand binding domains and intracellular kinase domains and the association with rapid Ca^{2+} activation. Nonetheless, because of the wave propagation and penetrability of temperature, the definition of thermosensors may not be limited to the plasma membrane. Temperature-triggered structural changes at the protein and RNA levels appear to be more pronounced for direct perception of temperature increases.

However, these identified thermosensors are mostly correlated with thermomorphogenesis, when plants encounter moderate increases in temperature. Whether similar mechanisms exist in HS perception remains to be further studied. Heat signal transduction is also achieved by means of subcellular translocation, especially translocation from the membrane to the nucleus. Nevertheless, the underlying mechanism that facilitates the translocation remains largely unknown. Splicing, anchoring by lipids, and chaperones are all possible ways to explain the *trans*-organellar translocation, which determines the definition of thermosensors. Further investigations based on these findings might provide new clues about the temperature-sensing mechanisms in plants.

We also propose multiple strategies to guarantee or even improve heat resilience in crops. Only understanding the network of heat responses from an overall perspective can help crop breeders address the global food security issues caused by global warming. The duration and magnitude of temperature elevation, different tissue temperatures and HS recovery should also be paid attention when exploring the mechanisms of heat responses in future research (Jagadish et al., 2021). In addition, the concept of a multifactorial stress combination should be taken into consideration in further investigations (Zandalinas et al., 2021). Environmental stresses always occur as combinations of two to three abiotic and/or biotic stresses. HS appears with drought stress, plant defense is compromised by increased temperature, and insect larvae have evolved to increase cuticular proteins to acquire heat acclimation in response to global warming. Therefore, the direction of future breeding programs should not be limited by focusing on individual environmental stresses (Kim et al., 2022a; Guo et al., 2022). Mining moonlighting regulators that are simultaneously involved in regulating thermotolerance, drought, disease, and insect resistance will resolve post-Green Revolution challenges in agriculture and identify emerging strategies to improve sustainable crop production and acclimation in the era of changing climate.

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